



Getting Most of ArgoCD

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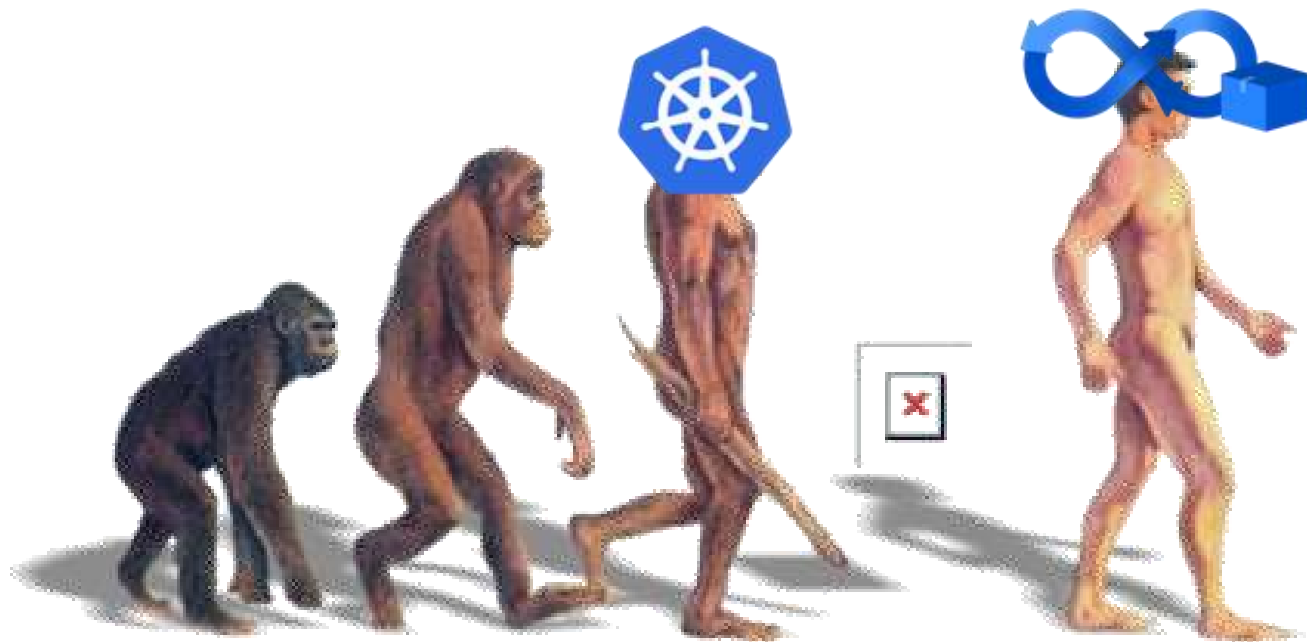


Cwise

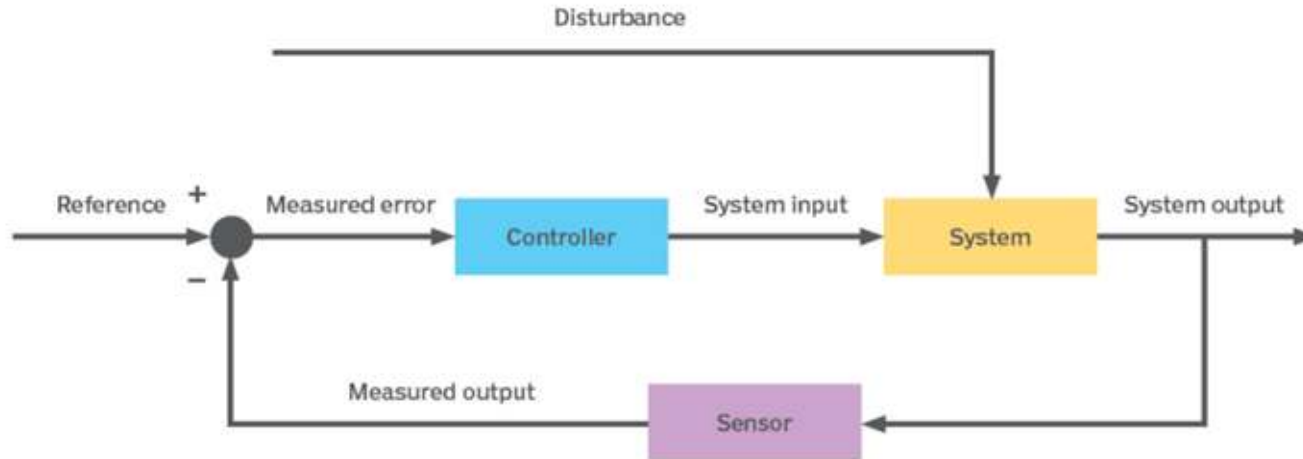
The Kubernetes



The Missing Link



Closed loop controller





gitops



ArgoCD



About Author

- **Doing DevOps Since 2013**
- **Mostly working with fintech/Financial Services**

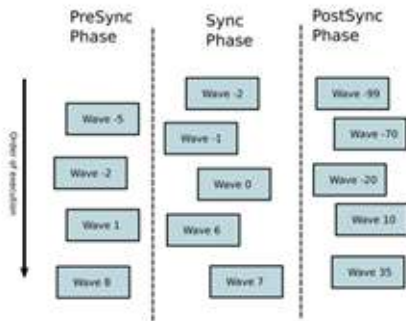


ArgoCD Techniques

Issue: **Applications
don't start in correct
order**

Organizing ArgoCD Start Ordering

Sync Waves and Phases:



Init Containers: Port/Connection

```
- name: wait-backend
  image: busybox
  command:
    - sh
    - -c
    - |
      echo -e "Check"
      while ! nc -z target 443; do
        sleep 1
        printf "- "
      done
      echo -e " >> Backend Available"
```

Init Containers: Wait for PlaceHolder File

```
- name: wait-for-migrations
  image: "busybox"
  command:
    - /bin/sh
    - /scripts/wait-for.sh
    - {{ .Values.app.imageTag | quote }}
    - {{ $mode | quote }}
```

Making Systems More
Discoverable

Issue: **Reducing Cognitive Load**

Making Systems More Discoverable

ArgoCD External URL's

annotations:

```
link.argocd.argoproj.io/external-link: http://my-grafana.example.com/pre-generated-link
```

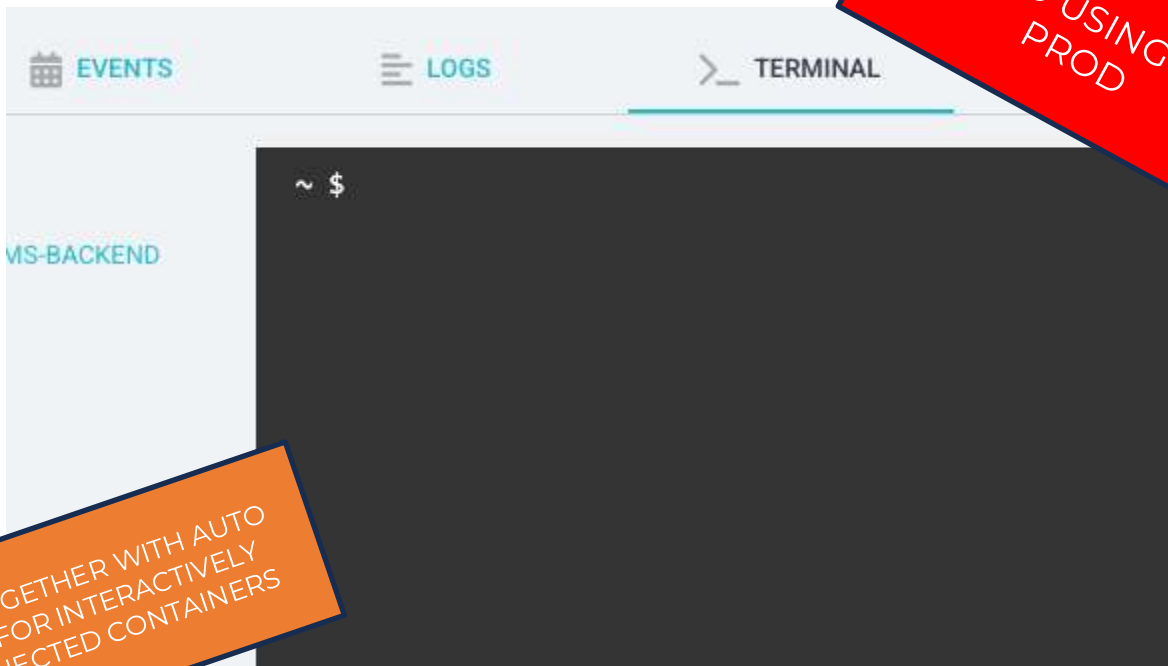


Temp Access To
System

Issue: **Access to systems**

Temp Access To
System

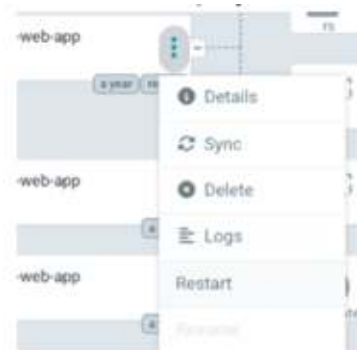
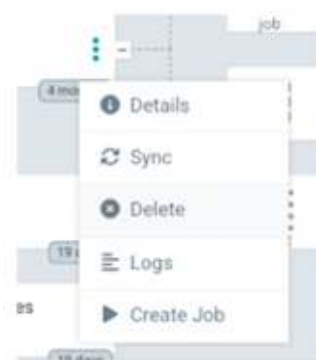
Web Terminal



Issue: **Provide Interface for common tasks**

Common Task Automation for Platform Clients

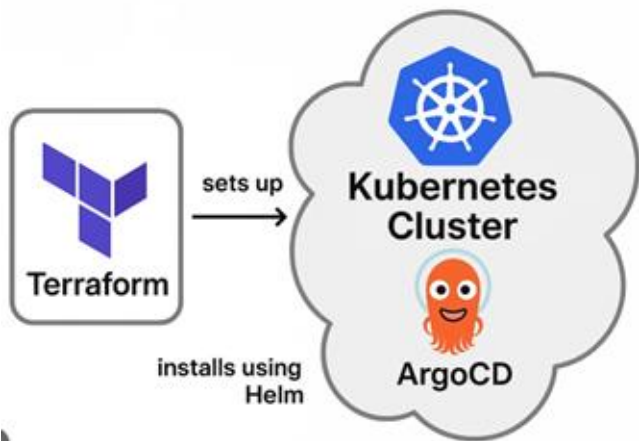
ArgoCD Resource Customizations with LUA



Issue: **More Argo's More Problems**

Config Management

Automated Deployment



ArgoCD Configs

```
resource "kubernetes_secret" "argoproj-https-creds-github-template" {  
  metadata {  
    name      = "argoproj-https-creds-bitbucket-template"  
    namespace = var.argocd_namespace  
    labels = {  
      "argocd.argoproj.io/secret-type" = "repo-creds"  
    }  
  }  
  data = {  
    url      = "https://bitbucket.org/example"  
    type     = "git"  
    username = var.argocd_bitbucket_repo_creds_user  
    password = var.argocd_bitbucket_repo_creds_pass  
  }  
}
```

Issue: **Creation App of Apps**

Deploying App-of-Apps from IaC

```
resource "kubernetes_manifest" "argo_application" {
  count = var.argo_cd_create_application ? 1 : 0
  manifest = {
    apiVersion = "argoproj.io/v1alpha1"
    kind       = "Application"
    metadata = {
      name      = "${var.country}-${var.brand}-${var.env}-app-of-apps"
      namespace = "argocd"
      labels    = local.common_tags
    }
  }
  spec = {
    destination = {
      namespace = kubernetes_namespace.ns.metadata.0.name
      server    = "https://kubernetes.default.svc"
    }
    project = "default"
    source = {
      path          = "${var.env}/"
      repoURL       = "https://github.com/example/${var.argo_cd_configuration_repo_name}.git"
      targetRevision = "HEAD"
    }
    syncPolicy = {
      automated = {}
    }
  }
}
```

Monitoring

Issue: **Detecting Failures**

Detecting Failures

Monitoring ArgoCD via Exporter



- am APP

Jan 26th at 1:11 PM

Alert: ArgoCD Degraded App Alert

Alert: ArgoCDDegradedApp | **Severity:** critical

Description: More than 0 ArgoCD applications found degraded here:



Thank You Very Much!



Cwise